

ENV 6106

Air Pollution Modeling

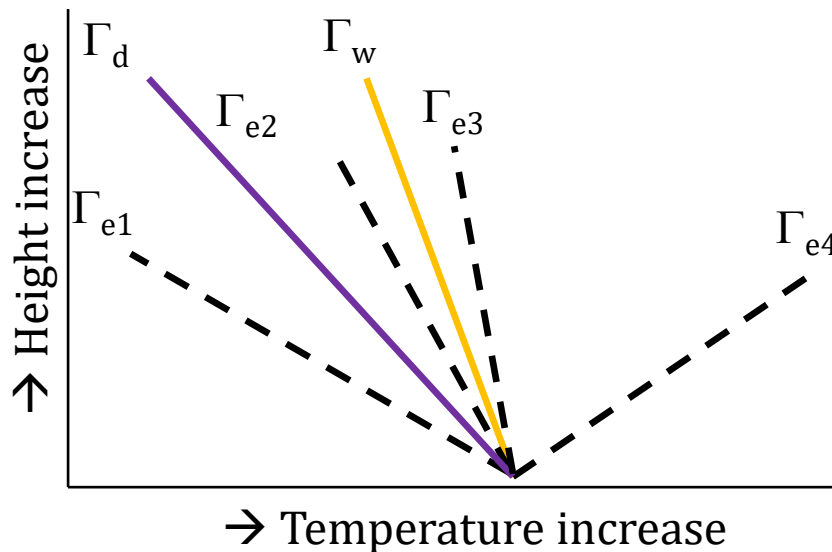
Spring 2023

Topics

- Gaussian dispersion model basics
- Expectation
 - Understand the concept behind Gaussian dispersion models
 - Be able to apply Gaussian models to estimate pollutant concentrations
- Readings: Cooper & Alley Chapter 20.1-20.3

Quick refresh of stability class

- ☐ $\Gamma_e > \Gamma_d$: Absolutely unstable (Γ_{e1})
- ☐ $\Gamma_e = \Gamma_d$: Dry neutral
- ☐ $\Gamma_d > \Gamma_e > \Gamma_w$: Conditionally unstable (Γ_{e2})
- ☐ $\Gamma_e = \Gamma_w$: Wet neutral
- ☐ $\Gamma_e < \Gamma_w$: Absolutely stable (Γ_{e3}, Γ_{e4})
- Real parcel lapse rate (Γ_p) generally between Γ_d and Γ_w .



Categorizing Stability: Classes

day: more stable as u increases
night: less stable as u increases

more stable with less sunlight

more stable with less clouds

Surface Wind				Night-Time Cloudiness	
(measured at 10 m)	Day-Time Insolation			Thinly Overcast or $\geq 4/8$ Cloudiness	
m s^{-1}	Strong	Moderate	Slight		$\leq 3/8$ Cloudiness
<1	A	A-B	B	-	-
2-3	A-B	B	C	E	F
3-5	B	B-C	C	D	E
5-6	C	C-D	D	D	D
>6	C	D	D	D	D

A-extremely unstable, B-moderately unstable, C-slightly unstable, D-neutral, E-slightly stable, F-moderately stable

(Hanna et al., 1982, Handbook on Atmospheric Diffusion. U.S. Dept. of Energy).

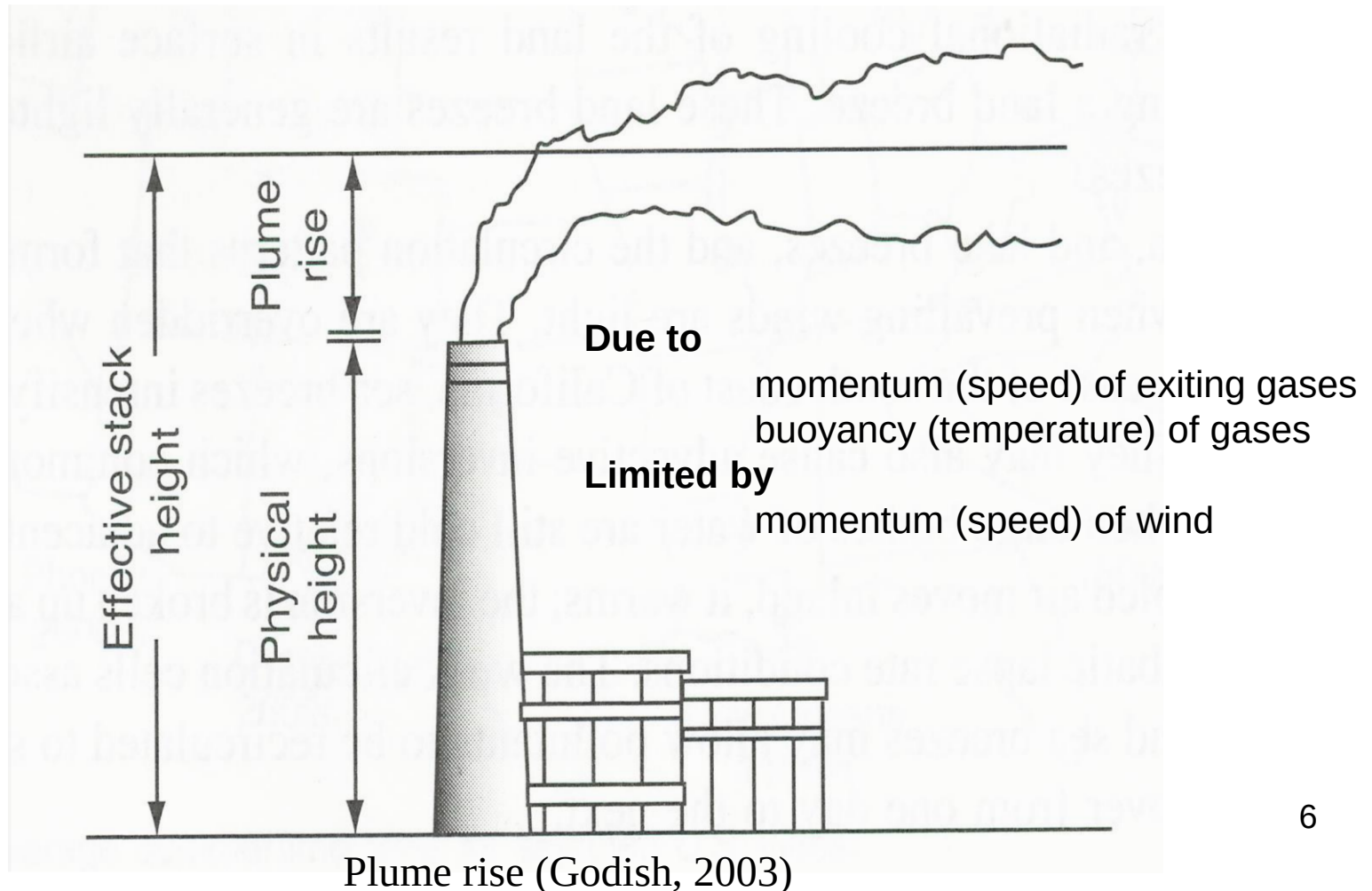
Motivation

- How to calculate pollutant concentrations at one receptor location due to emissions?
- Start with a simple stack....
- First understand the plume from the stack...

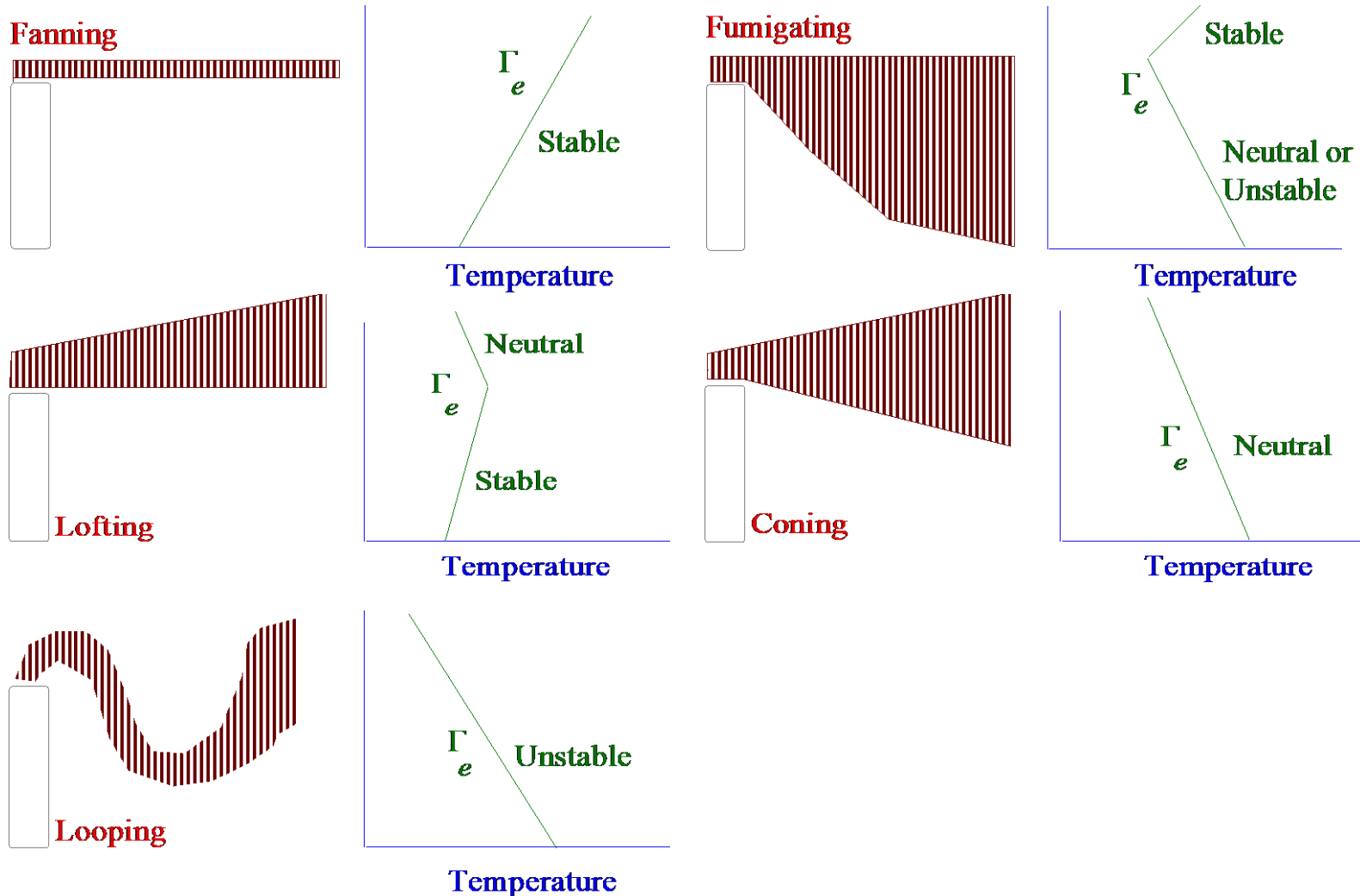


Plume rise

- Plume first rises after release

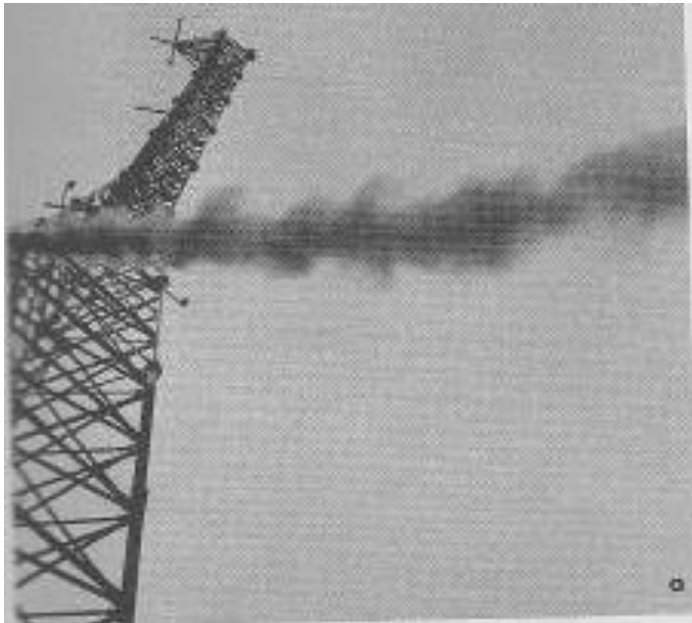


Plume shape vs stability



Time averaged plume

- Instantaneously, plume shape varies
- On average: “Coning” plume



1/25 seconds exposure

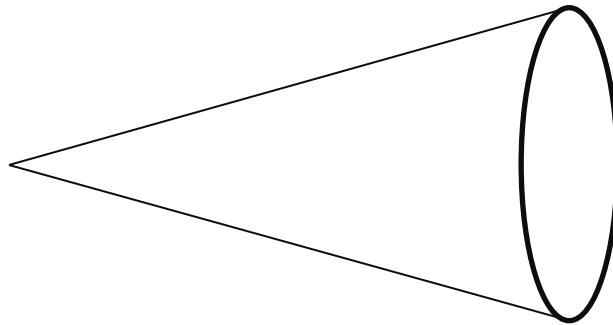


5 minutes exposure

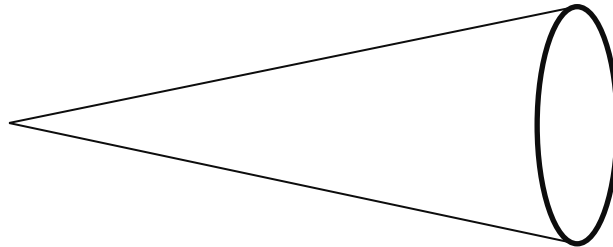
Time averaged plume

- More unstable atmosphere → more spread out

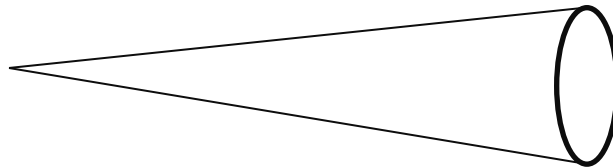
Unstable



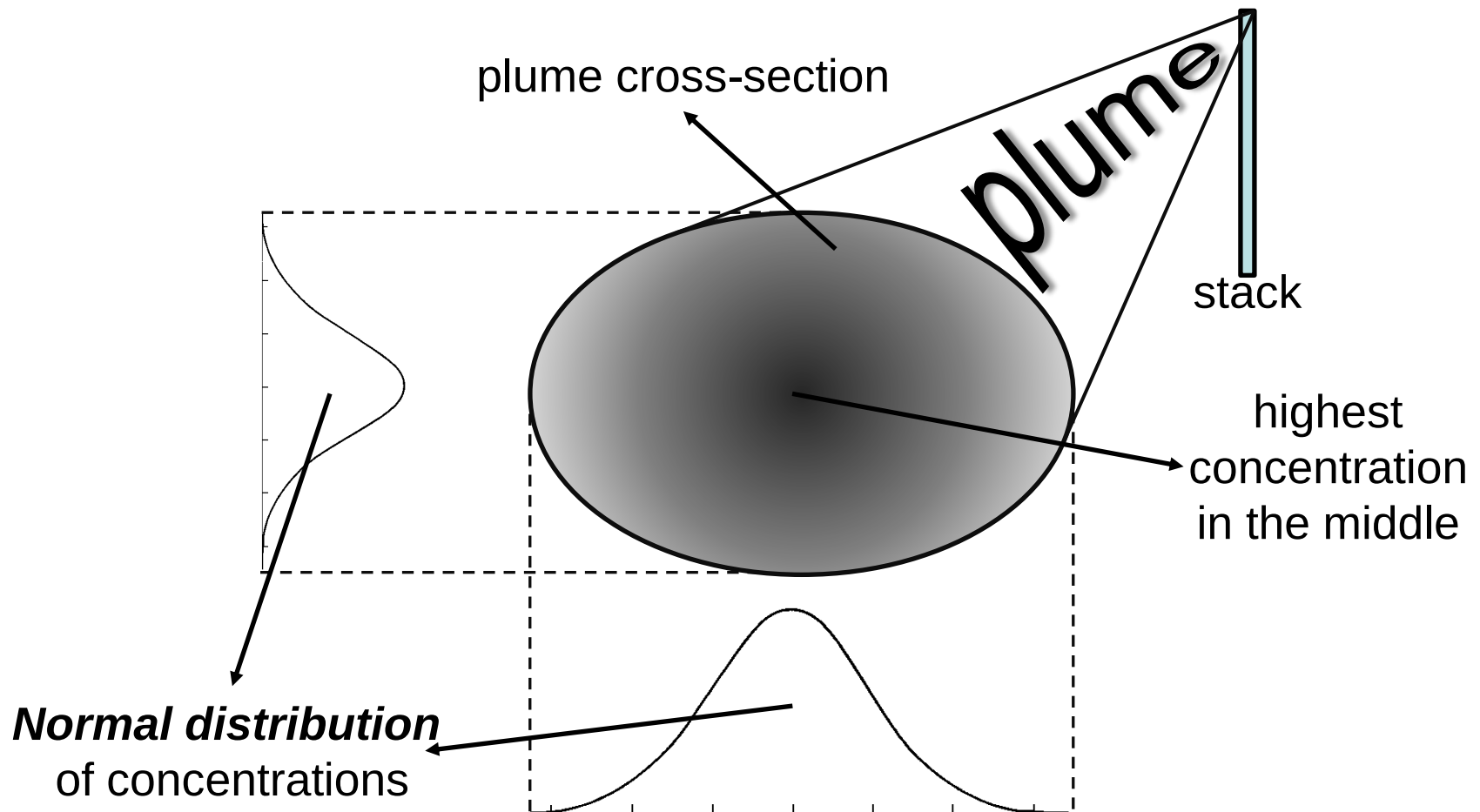
Neutral



Stable



Inside the time average plume...

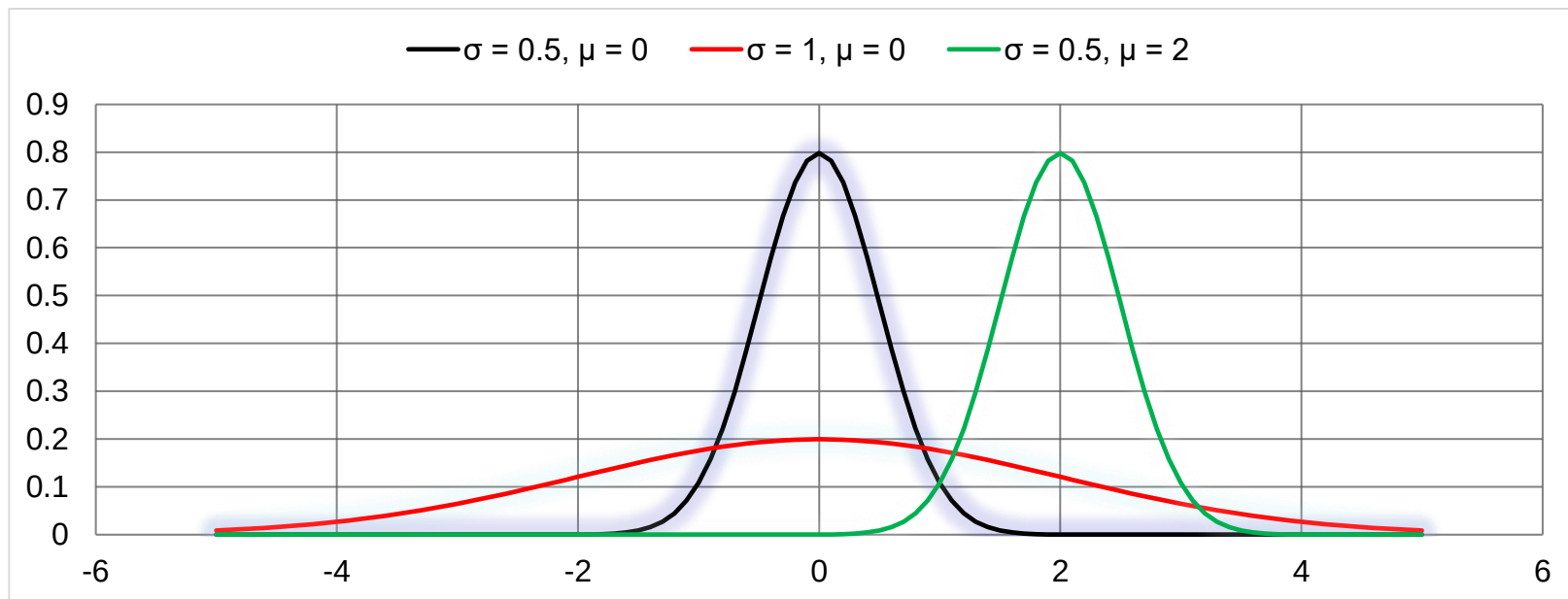


What is normal distribution

- Also called Gaussian distribution

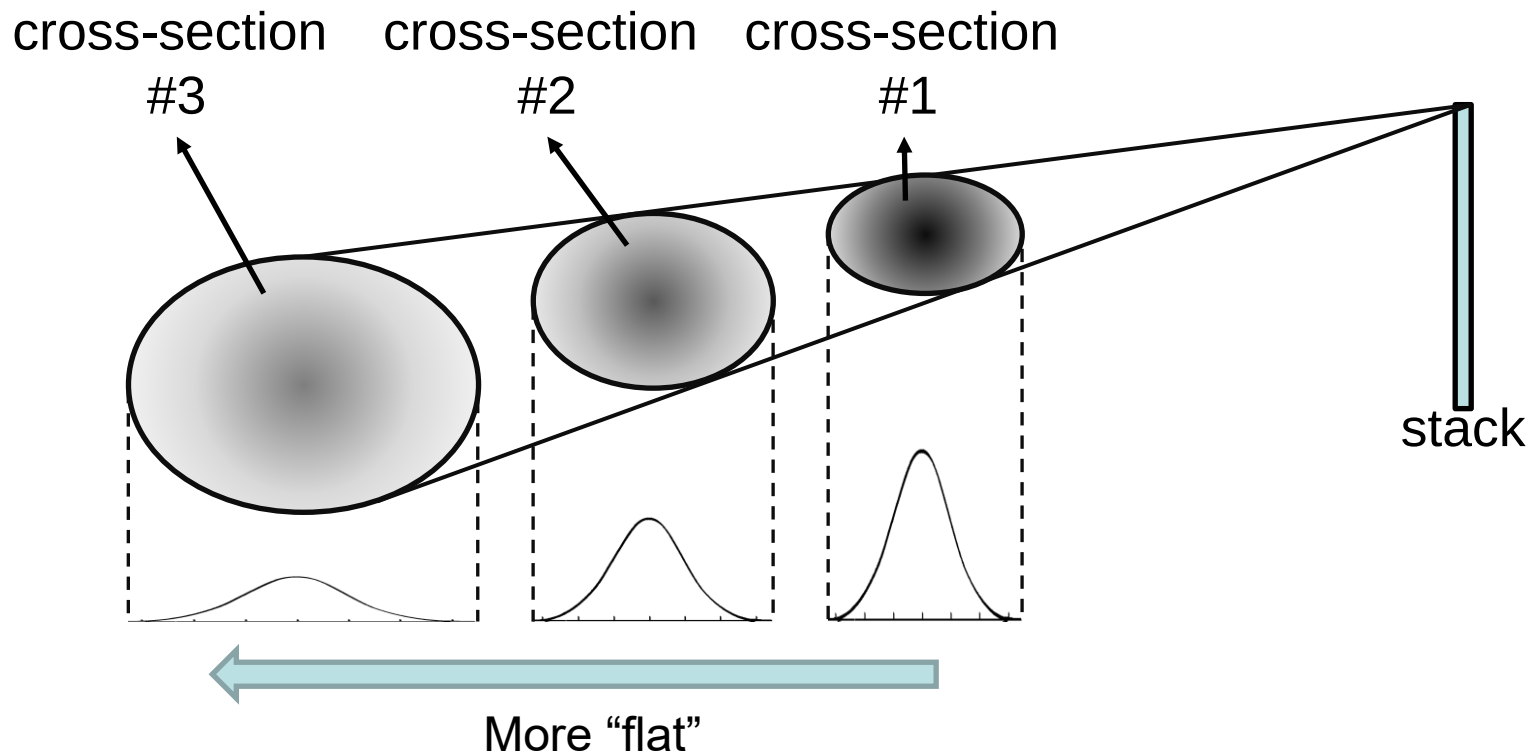
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Center of the distribution determined by μ
- “Flatness” of the distribution determined by σ

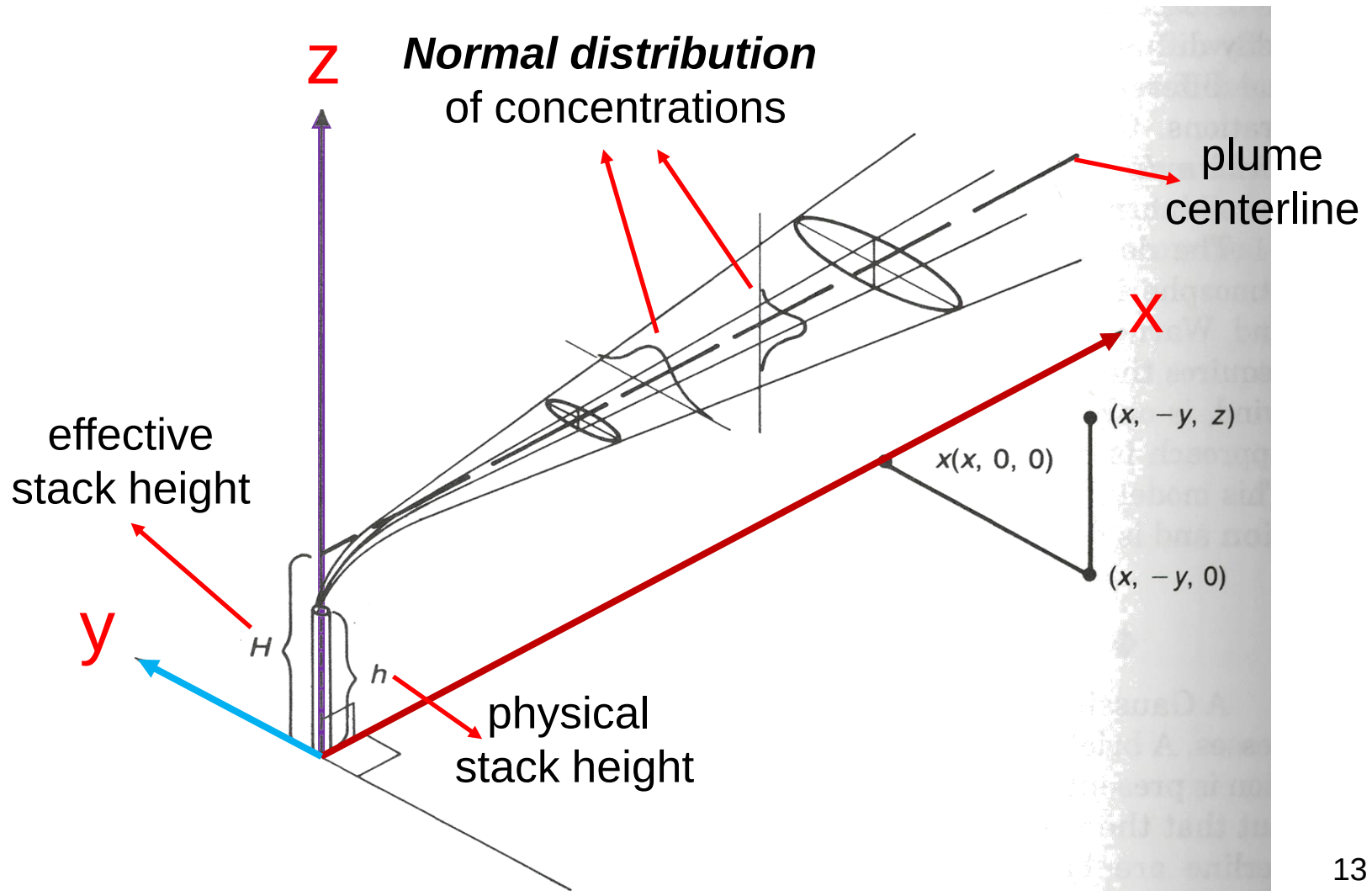


Inside the time average plume...

- The concentration distribution widens (increase σ) with increase in distance



Inside the time-average plume...



The Gaussian Plume Model

- Basic Gaussian Plume Model

Pollutant concentration at downwind location (x,y,z)

Pollutant emission rate

Effective height

$$C(x,y,z) = \frac{S}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left\{ \exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) \right\}$$

Wind speed at effective height H

Gaussian distribution parameter, determine "flatness" at y (σ_y) and z (σ_z) direction

The Gaussian Plume Model

- Take account into ground reflection

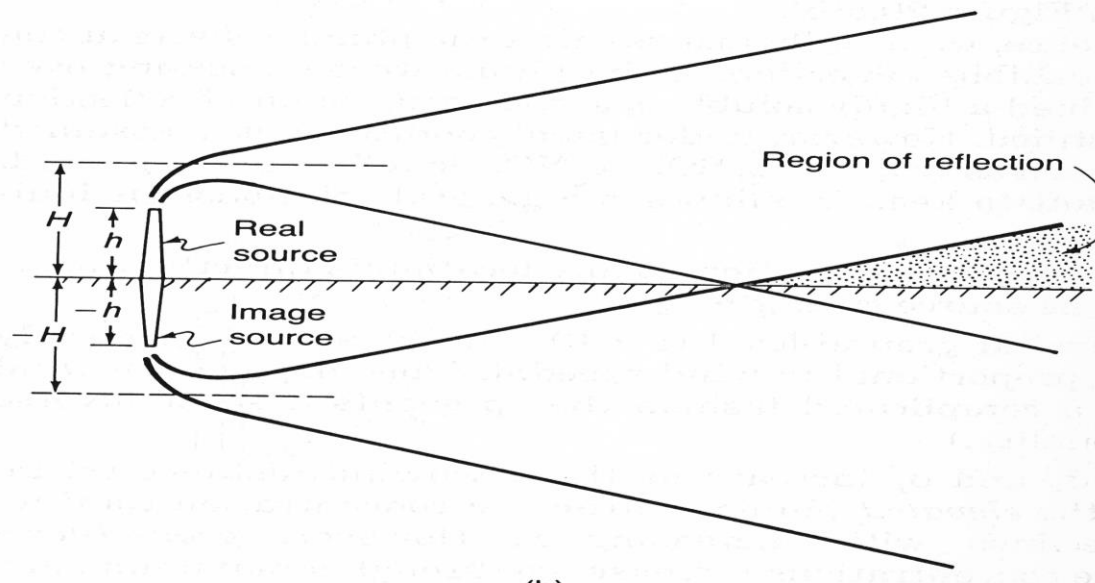


Figure 20.5 (Cooper and Alley, 2002)

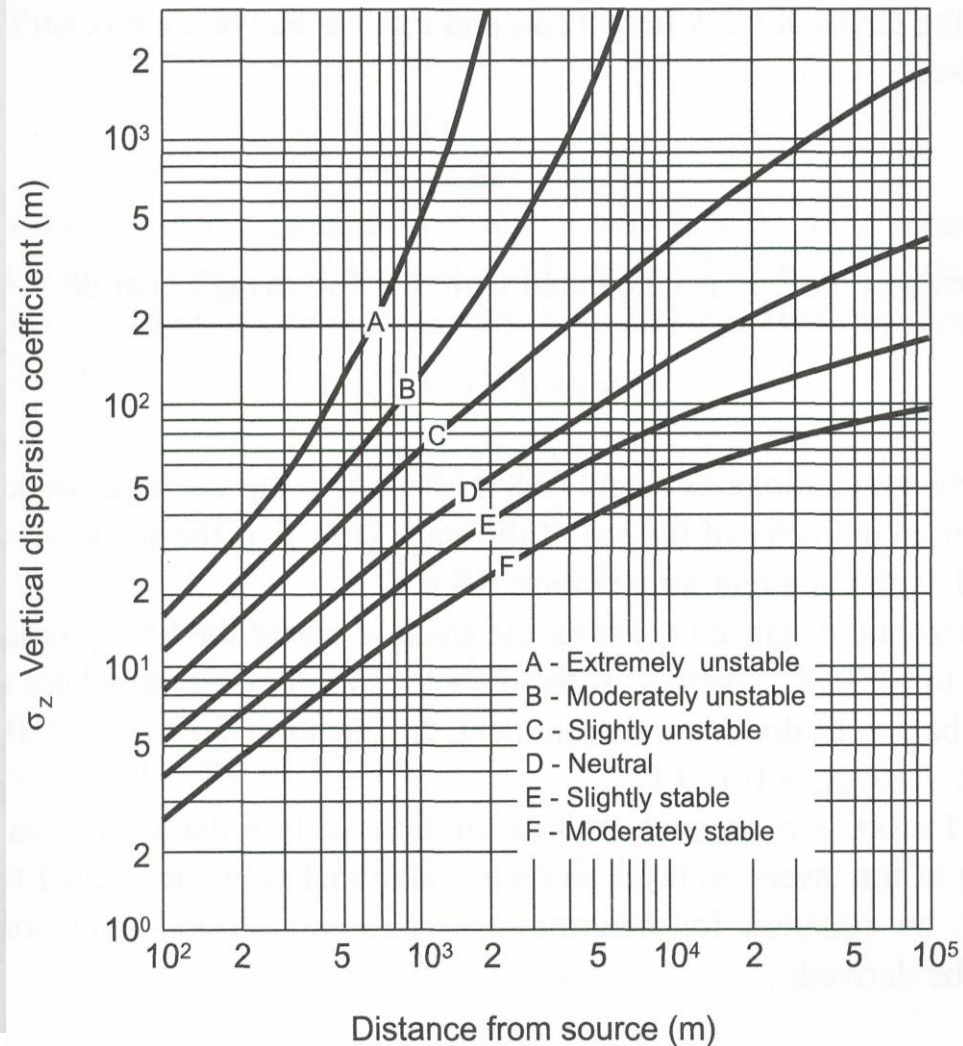
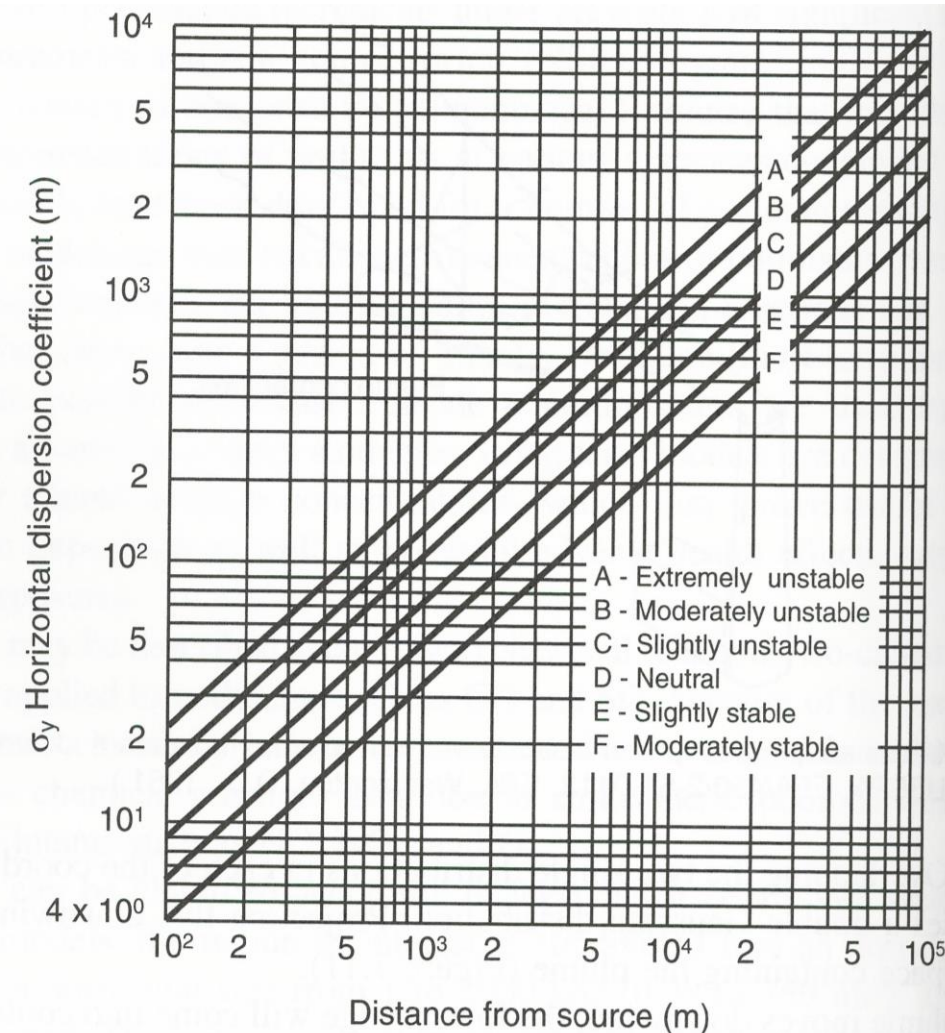
$$C(x,y,z) = \frac{S}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left\{ \exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right) \right\}$$

*This one is the **most used** form of Gaussian plume model*

How to find σ_y and σ_z

- Two methods to find σ_y and σ_z
 - Look them up in two figures
 - Calculate them using fitted equations
 - There are many different fitted equations available
 - Based on stability class and distance

How to find σ_y and σ_z (*method 1*)



(Godish, 2003)

How to find σ_y and σ_z (*method 2*)

Table 20.2 Values of Curve-Fit Constants for Calculating Dispersion Coefficients as a Function of Downwind Distance and Atmospheric Stability

Stability	<i>a</i>	<i>b</i>	<i>x</i> < 1 km			<i>x</i> > 1 km		
			<i>c</i>	<i>d</i>	<i>f</i>	<i>c</i>	<i>d</i>	<i>f</i>
A	213	0.894	440.8	1.941	9.27	459.7	2.094	-9.6
B	156	0.894	106.6	1.149	3.3	108.2	1.098	2.0
C	104	0.894	61.0	0.911	0	61.0	0.911	0
D	68	0.894	33.2	0.725	-1.7	44.5	0.516	-13.0
E	50.5	0.894	22.8	0.678	-1.3	55.4	0.305	-34.0
F	34	0.894	14.35	0.740	-0.35	62.6	0.180	-48.6

Adapted from Martin, 1976.

$$\sigma_y(x) = ax^b$$

$$\sigma_z(x) = cx^d + f$$

$$\sigma [=] \text{ m}, x [=] \text{ km}$$

Assumptions of Gaussian Plume Model

- Steady-state conditions
 - Constant wind direction, speed etc..
 - No spatial variation of wind field...
 - No background concentrations
- No terrain effect.
 - More assumptions see Abdel-Rahman, 2008 (p32)
- Mainly applied in short range ($< 50\text{km}$), flat terrain, simple meteorological conditions